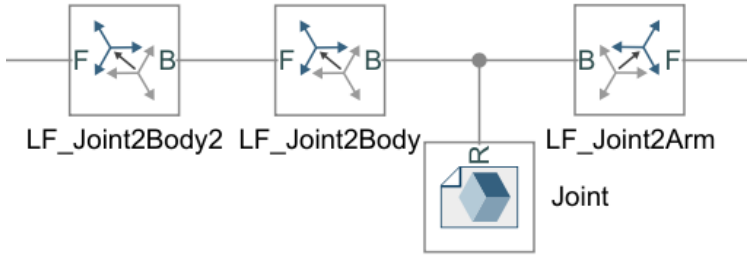
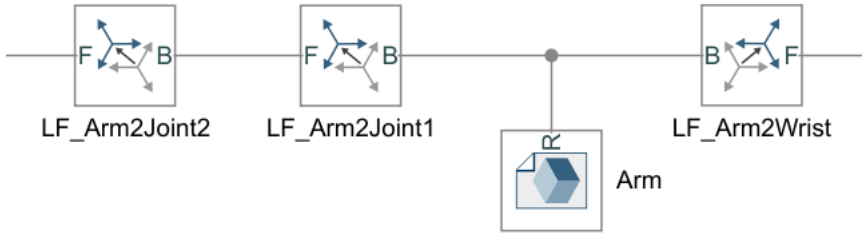
机器狗Simulink仿真及正运动学分析

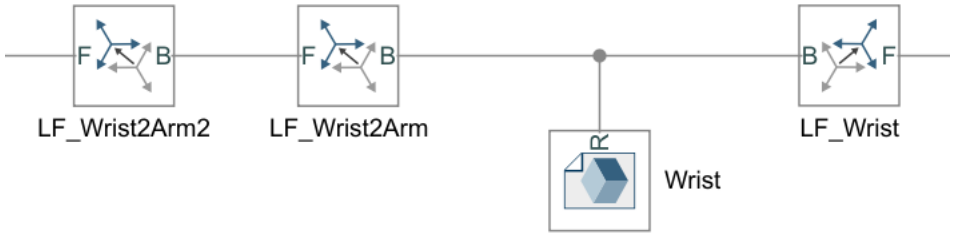
侯红日PB19050979

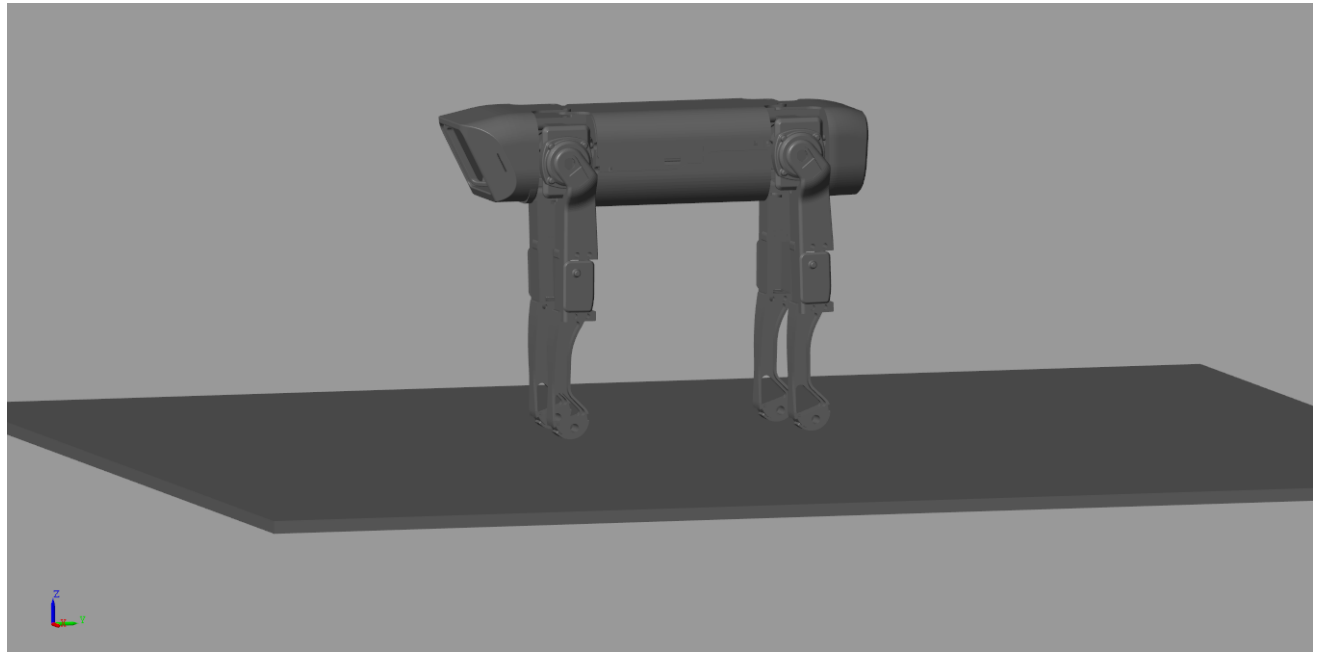
20221025

Simulink导入各部件



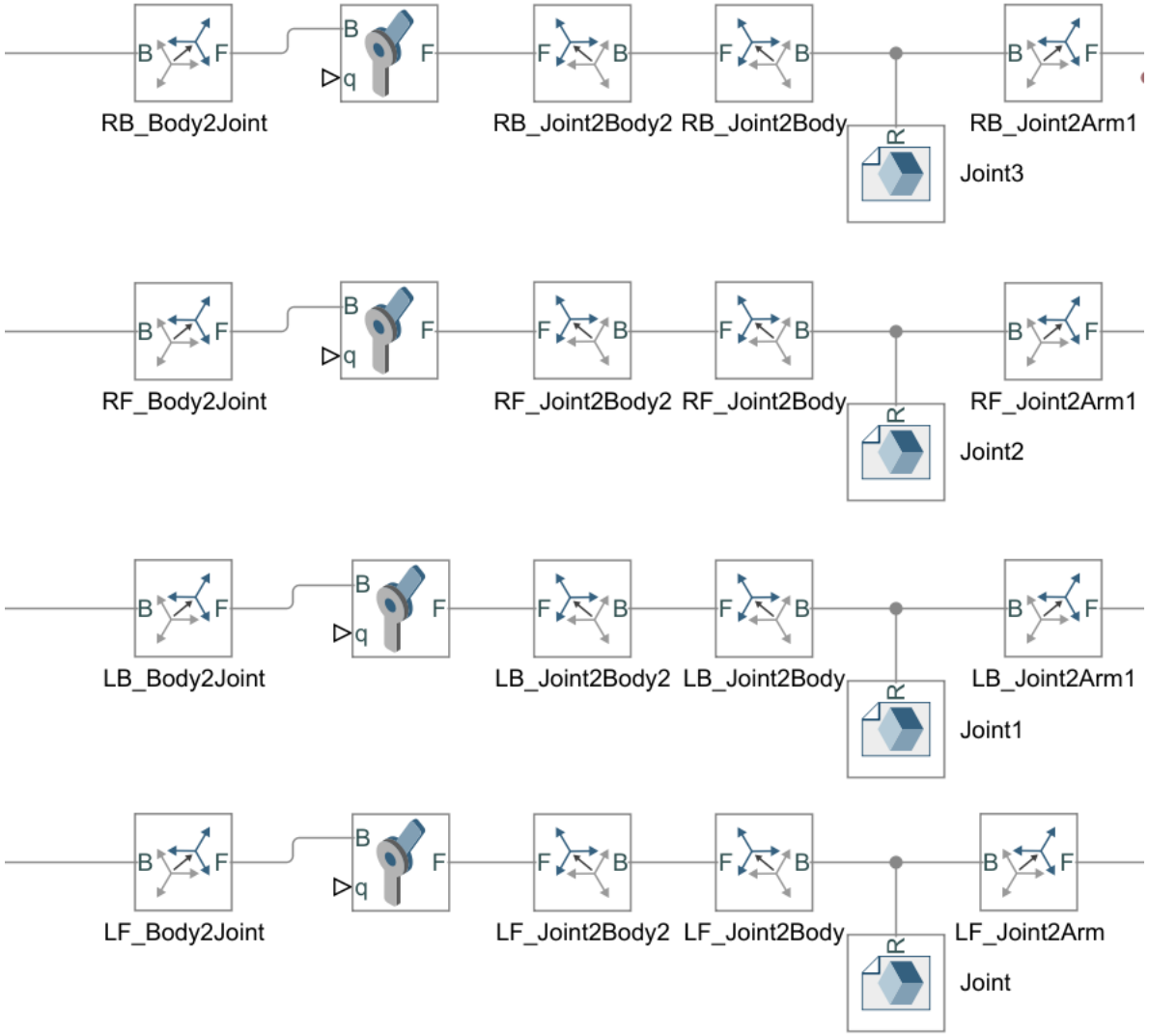






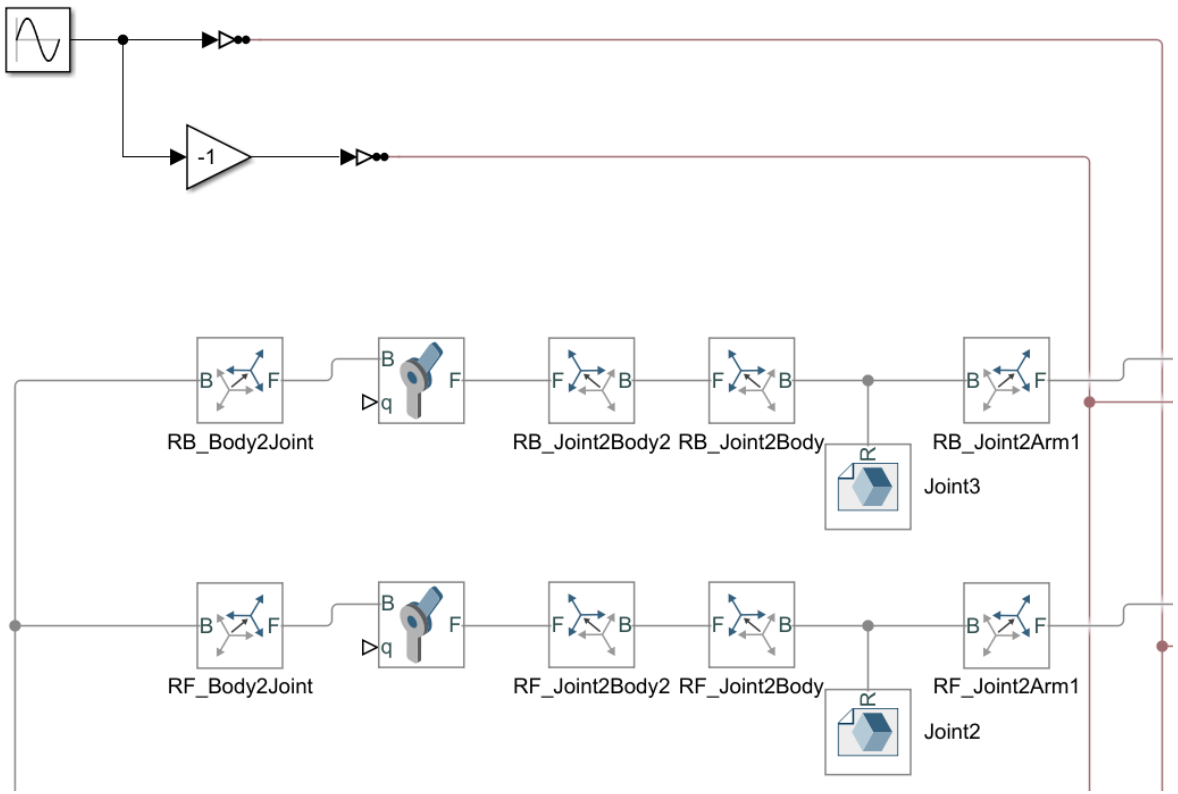
Simulink导入身体和腿部的三个构件并进行编译仿真，从而将模型导入Simulink

Simulink构建运动副



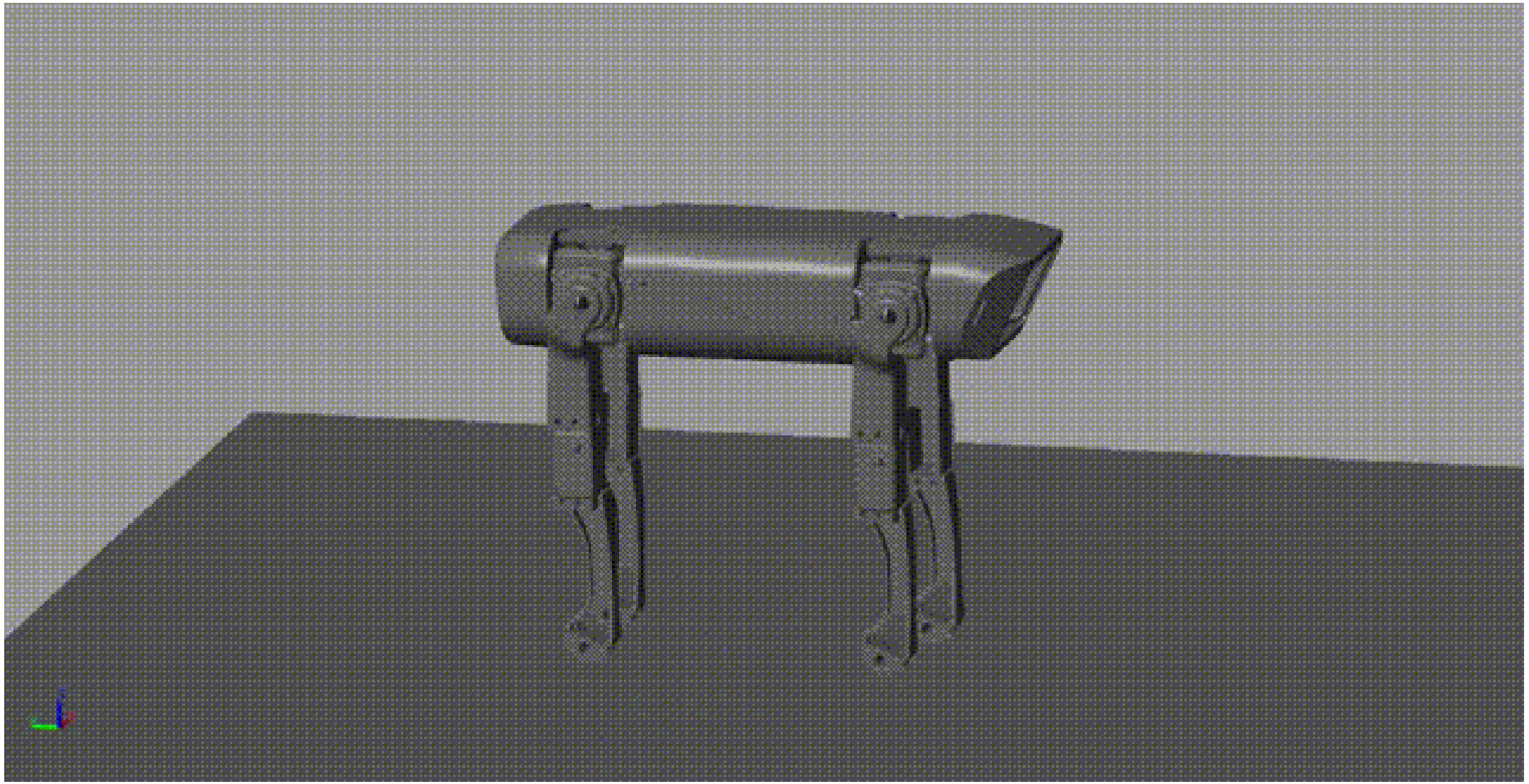
导入三个腿部构件与身体互相之间的连接点的运动副来构建运动关系，否则为刚性连接

Simulink简单运动仿真

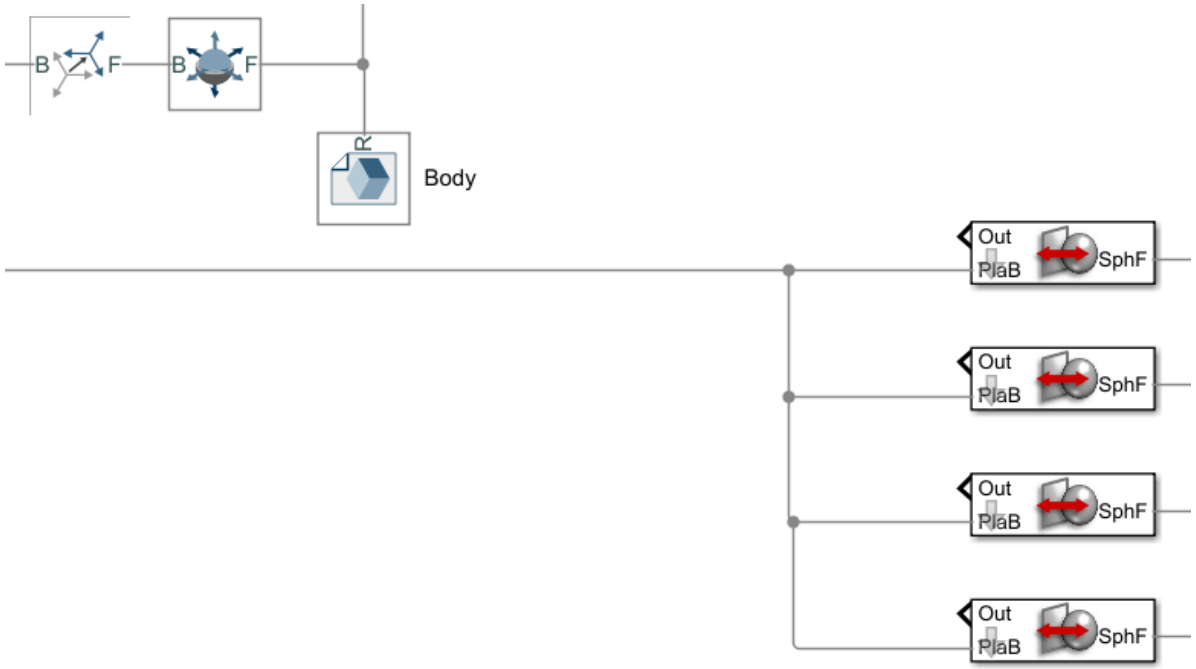


设置运动的波形发生器，使得四条腿可以运动；此处仅设置了髋关节和大腿之间的运动副运动关系，步态规划部分再详细规划

Simulink简单运动仿真

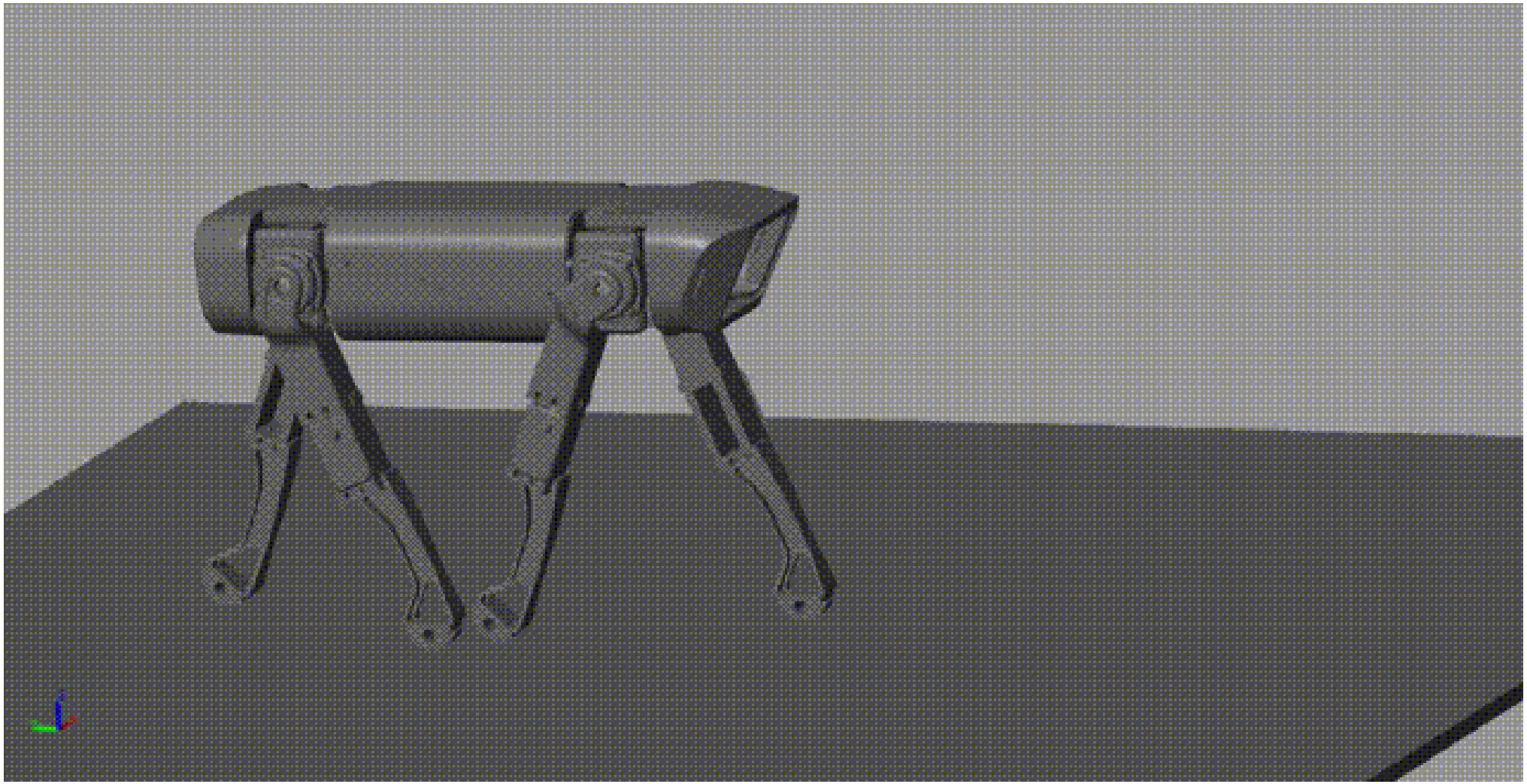


Simulink力学反馈

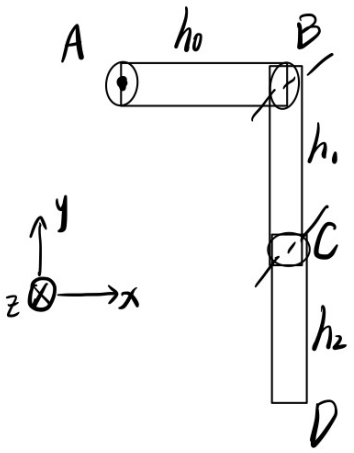


确定力学反馈，使得身体与平面之间存在力学反馈，以及四条腿的末端与平面之间的力学反馈

Simulink力学反馈



正运动学分析

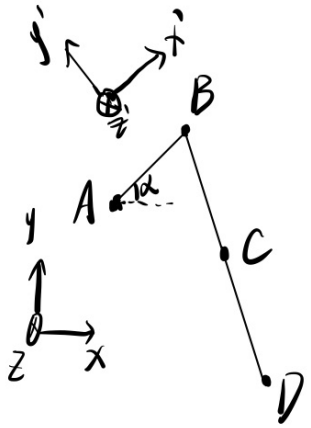


$$\begin{pmatrix} x_D \\ y_D \\ z_D \end{pmatrix} = \begin{pmatrix} D_{11} & D_{12} & D_{13} \\ D_{21} & D_{22} & D_{23} \\ D_{31} & D_{32} & D_{33} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

$A \to B \to C, D$

$A(0, 0, 0)$

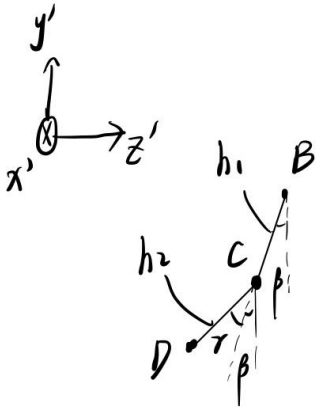
B:
$$\begin{pmatrix} x_B \\ y_B \\ z_B \end{pmatrix} = \begin{pmatrix} h_0 \cos\alpha \\ h_0 \sin\alpha \\ 0 \end{pmatrix}$$



对于 $x'y'z'$ 坐标系 → xyz 坐标系的变换.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + \begin{pmatrix} x_B \\ y_B \\ z_B \end{pmatrix}$$

正运动学分析



在 $x'y'z'$ 坐标系下

$$\begin{pmatrix} x_D \\ y_D \\ z_D \end{pmatrix} = \begin{pmatrix} x_B \\ -h_1\sin\beta - h_2\sin(\beta+\gamma) \\ -h_1\cos\beta - h_2\cos(\beta+\gamma) \end{pmatrix}$$

代入坐标变换式即可得到 D点在 xyz 坐标系下的坐标

在 xyz 坐标系下,

$$\begin{pmatrix} x_D \\ y_D \\ z_D \end{pmatrix} = \begin{pmatrix} h_0\cos^2\alpha + h_1\sin\alpha\sin\beta + h_2\sin(\beta+\gamma) + h_0\cos\alpha \\ h_0\sin\alpha\cos\alpha - h_1\cos\alpha\sin\beta - h_2\cos\alpha\sin(\beta+\gamma) + h_1\sin\alpha \\ -h_1\cos\beta - h_2\cos(\beta+\gamma) \end{pmatrix}$$

后期工作

1. 进行逆运动学分析并用程序实现，进行步态规划方面的调研；
2. 使用Matlab的Simulink模块进行步态规划；
3. 落实机械设计方面的图纸规划；
4. 进行测量算法等方面的调研及应用。